Package Reference

LaTeX packages and commands

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1 Code display packages

1.1 listings – Source code

Template setup

```
\input{Packages/Code/usepackages}
```

The template defines `matlab` and `latex` in `Packages/Code/package_config.tex`. Use those styles directly.

MATLAB code

```
\begin{lstlisting}[style=matlab]
x = linspace(0, 2*pi, 100);
y = sin(x);
plot(x, y)
xlabel('x')
ylabel('sin(x)')
\end{lstlisting}
```

```
x = linspace(0, 2*pi, 100);
y = sin(x);
plot(x, y)
xlabel('x')
ylabel('sin(x)')
```

LaTeX code

```
\begin{lstlisting}[style=latex]
\begin{equation}
    F = ma
\end{equation}
\end{lstlisting}
```

```
\begin{equation}
    F = ma
\end{equation}
```

Read code from a file

The following script exists in this repository. This example displays its first seven lines; remove the line limits to display the complete file.

```
\lstinputlisting[style=matlab,firstline=1,lastline=7]
    {../MATLAB-scripts/initialize_plot_configuration.m}
```

```
function initialize_plot_configuration()
% set the size of the ticks on the axes
set(groot , 'defaultLegendFontSize' , 12);
% set the default size of the text
set(groot , 'defaultTextFontSize' , 12);
% set the default axes font size.
set(groot , 'defaultAxesFontSize' , 12);
```

1.2 xcolor – Colours

Preamble

```
\usepackage{xcolor}
\definecolor{accent}{gray}{0.35}
```

Local colour

```
\textcolor{accent}{Text in dark grey.}
{\color{accent}A local colour change.}
```

Text in dark grey. A local colour change.

`\textcolor` affects its argument; `\color` affects the rest of the current group. Define colours once in the preamble.

2 Graphics packages

2.1 booktabs – Tables

Preamble

```
\usepackage{booktabs,siunitx}
```

Table with aligned numbers

```
\begin{table}[htbp]
    \centering
    \caption{Test results.}
    \label{tab:tests}
    \begin{tabular}{l S[table-format=2.2]}
        \toprule
        Test & {Time (\unit{\second})} \\
        \midrule
        A & 9.25 \\
        B & 12.40 \\
        C & 8.70 \\
        \bottomrule
    \end{tabular}
\end{table}
```

Table 1: Test results.

Test	Time (s)
A	9.25
B	12.40
C	8.70

`S[table-format=2.2]` (from `siunitx`) reserves two integer digits and two decimals. Use `tabularx` for text columns and `longtable` for tables spanning pages.

2.2 graphicx – Images

Preamble

```
\usepackage{graphicx}
```

Single image

```
\begin{figure}[htbp]
    \centering
    \includegraphics[width=0.4\linewidth]
        {Images/image_placeholder.png}
    \caption{System response to a step input.}
    \label{fig:response}
\end{figure}

See figure~\ref{fig:response}.
```

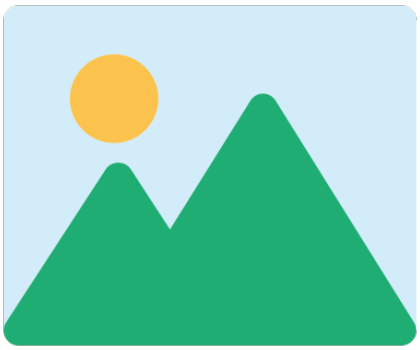


Figure 1: System response to a step input.

See figure 1.

Size limits without distortion

```
\includegraphics[
    width=0.8\linewidth,
    height=0.35\textheight,
    keepaspectratio
]{Images/image_placeholder.png}
```

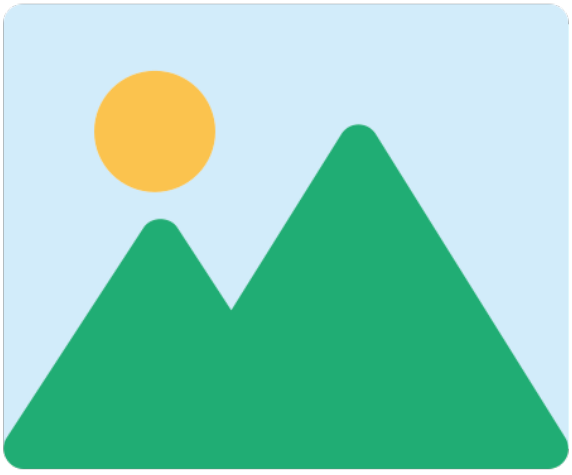


Figure 2: Image constrained by width and height.

Use `\linewidth` inside columns or minipages. `[htbp]` allows floating; `[H]` fixes the image in place and requires `float`. Put `\label` after `\caption`.

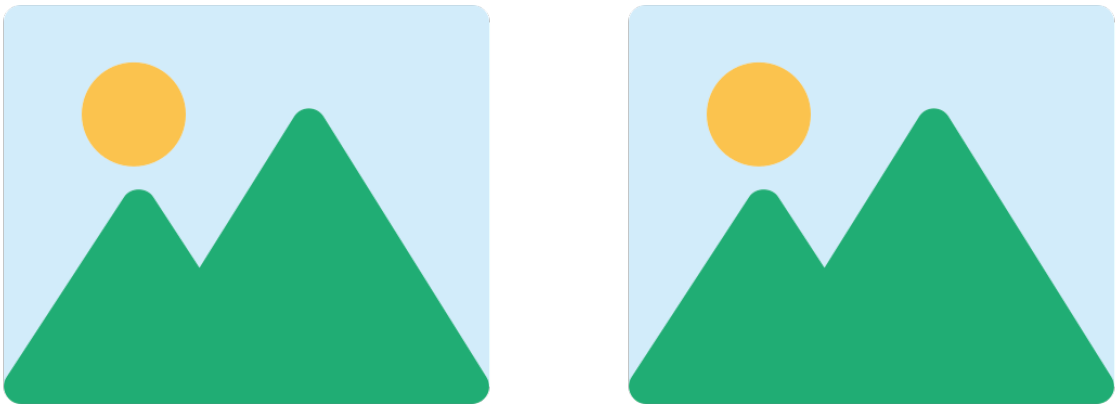
2.3 subcaption – Side-by-side images

Preamble

```
\usepackage{graphicx,subcaption}
```

Two panels with one figure number

```
\begin{figure}[htbp]
    \centering
    \begin{subfigure}{0.47\linewidth}
        \centering
        \includegraphics[width=\linewidth]
            {Images/image_placeholder.png}
        \caption{Without a controller.}
        \label{fig:uncontrolled}
    \end{subfigure}\hfill
    \begin{subfigure}{0.47\linewidth}
        \centering
        \includegraphics[width=\linewidth]
            {Images/image_placeholder.png}
        \caption{With a controller.}
        \label{fig:controlled}
    \end{subfigure}
    \caption{Comparison of the time responses.}
    \label{fig:comparison}
\end{figure}
```



(a) Without a controller.

(b) With a controller.

Figure 3: Comparison of the time responses.

Use `\ref{fig:comparison}` for the complete figure and `\subref{fig:uncontrolled}` for a panel. Replace `subfigure` with `minipage` for independent figure numbers.

2.4 tikz – Block diagrams

Template setup – styles used by both examples

```
\input{Packages/Graphics/usepackages}
% Loads TikZ and the template styles from:
% Packages/Graphics/package_config.tex
```

Series blocks

```
\begin{tikzpicture}[node distance=12mm]
    \node[input] (r) {$R(s)$};
    \node[block,right=of r] (g1) {$s+4$};
    \node[block,right=of g1] (g2) {$G(s)$};
    \node[output,right=of g2] (y) {$Y(s)$};
    \draw[->] (r) -- (g1);
    \draw[->] (g1) -- (g2);
    \draw[->] (g2) -- (y);
\end{tikzpicture}
```



Figure 4: Two blocks in series.

`right=of g1` uses `node distance`; `below=15mm of g1` sets a local distance. `(g1.south)` selects the lower anchor. The template defines `input`, `output`, `block`, `sum`, `pickoff`, `plus` and `minus` in `Packages/Graphics/package_config.tex`.

Feedback loop

Uses the template styles listed on page 7.

```
\begin{tikzpicture}[node distance=11mm]
    \node[input] (r) {$R(s)$};
    \node[sum,right=of r] (s) {};
    \node[block,right=of s] (g) {$G(s)$};
    \node[pickoff,right=of g] (p) {};
    \node[output,right=of p] (y) {$Y(s)$};
    \coordinate (mid) at ($(s)!0.5!(p)$);
    \node[block,below=12mm of mid] (h) {$H(s)$};
    \draw[->] (r) -- node[near end,above] {$+$} (s);
    \draw[->] (s) -- (g);
    \draw (g) -- (p);
    \draw[->] (p) -- (y);
    \draw[->] (p) |- (h.east);
    \draw[->] (h.west) -| node[near end,left] {$-$} (s.south);
\end{tikzpicture}
```

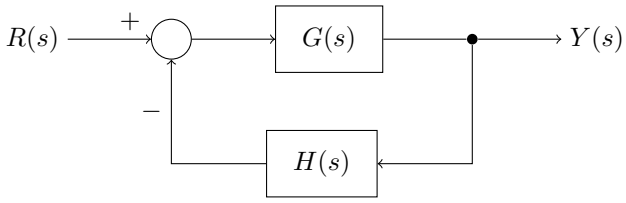


Figure 5: Negative-feedback block diagram.

Coordinates and scaling

```
\begin{tikzpicture}[scale=0.85,transform shape]
    \coordinate (s) at (0,0);
    \coordinate (p) at (8,0);
    \coordinate (quarter) at ($(s)!0.25!(p)$);
    \coordinate (centre) at ($(s)!0.50!(p)$);
    \coordinate (threequarters) at ($(s)!0.75!(p)$);
    \draw[->] (s) -- (p);
    \foreach \point/\percent in
        {s/0,quarter/25,centre/50,threequarters/75,p/100} {
        \fill (\point) circle (1.5pt);
        \node[below=3mm] at (\point) {\percent\%};
    }
\end{tikzpicture}
```



Figure 6: Interpolated coordinates at a scale of 0.85.

`--` draws a straight line; `|-` goes vertically then horizontally; `-|` does the reverse. `calc` enables interpolation between nodes. `transform shape` scales the node text and shapes too.

2.5 wrapfig – Text around an image

Preamble

```
\usepackage{graphicx,wrapfig,lipsum}
```

Image followed by a paragraph

```
\begin{wrapfigure}{r}{0.32\textwidth}
    \centering
    \includegraphics[width=\linewidth]
        {Images/image_placeholder.png}
    \caption{Experimental setup.}
    \label{fig:setup}
\end{wrapfigure}
\lipsum[1]
```

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

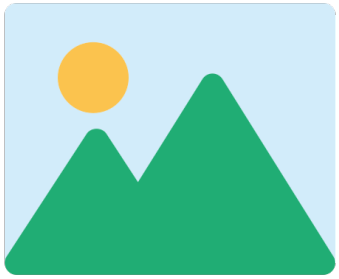


Figure 7: Experimental setup.

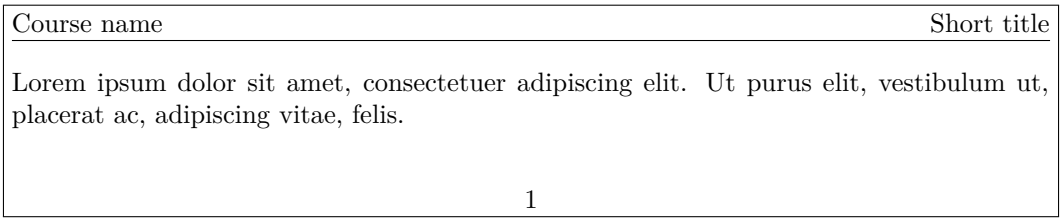
`r` places the image on the right; `l`, on the left. The optional argument `[10]` sets the number of wrapped lines. Avoid nearby headings, lists and page breaks.

3 Document layout packages

3.1 fancyhdr and geometry – Page layout

Preamble

```
\usepackage[margin=25mm,headheight=24pt]{geometry}
\usepackage{fancyhdr}
\pagestyle{fancy}
\fancyhf{}
\fancyhead[L]{Course name}
\fancyhead[R]{Short title}
\fancyfoot[C]{\thepage}
\setlength{\parindent}{0pt}
\setlength{\parskip}{6pt}
```



Header and footer shown at reduced page height. `geometry` sets the margins; `fancyhdr` sets the header and footer. For one page without either, use `\thispagestyle{empty}`. `\linewidth` is the local width; `\textwidth` is the full page text width.

3.2 hyperref – Links and references

Preamble

```
\usepackage{amsmath}
\usepackage[hidelinks]{hyperref}
```

Labels and references

```
\section{Model}
\label{sec:model}

\begin{equation}
    F = ma
    \label{eq:force}
\end{equation}

See section~\ref{sec:model}.
We use equation~\eqref{eq:force}.
\hyperref[sec:model]{Return to the model}.
\href{https://www.uc3m.es}{UC3M website}.
\url{https://www.uc3m.es}
```

1 Model

$$F = ma \tag{1}$$

See section 1. We use equation (1). Return to the model. UC3M website. `https://www.uc3m.es`

Unnumbered heading in the contents

```
\tableofcontents
% An unnumbered section included in the contents:
\section*{Acknowledgements}
\phantomsection
\addcontentsline{toc}{section}{Acknowledgements}
```

Contents

Acknowledgements 1

Acknowledgements

The heading stays unnumbered while its entry appears in the contents. Compile twice to resolve references. Figure and table labels go after their captions; equation labels go inside the equation environment.

3.3 multicol – Multiple columns

Preamble

```
\usepackage{multicol,lipsum}
\setlength{\columnsep}{8mm}
\setlength{\columnseprule}{0pt}
```

Balanced columns

```
\begin{multicols}{2}
    \lipsum[1][1-4]
    \columnbreak
    \lipsum[2][1-4]
\end{multicols}
```

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi.

Use `\linewidth` to size content inside a column. Ordinary single-column floats are not supported within `multicols`.

Columns inside a box – balanced

```
\begin{multicols}{3}
    \lipsum[1][1-2]
    \columnbreak
    \lipsum[2][1-2]
    \columnbreak
    \lipsum[3][1-2]
\end{multicols}
```

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero.

3.4 needspace – Page break control

For keeping a complete short block together, use the template's `nobreaks` environment on page 12. `needspace` remains useful when only the beginning of a longer section needs to stay with its heading.

Preamble

```
\usepackage{needspace,lipsum}
```

Reserve room before a heading

```
\lipsum[1][1-3]

\Needspace{8\baselineskip}
\subsection{Results}
\lipsum[2][1-4]
```

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris.

Results

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi.

If fewer than eight baseline spacings remain, the heading and its opening text start on the next page.

Command	Use
`\Needspace{3cm}`	Break if less than 3 cm remains.
`\needspace{3cm}`	Approximate reservation.
`\Needspace*{3cm}`	Preserve bottom alignment with `\flushbottom`.
`\newpage`	Force a page break.
`\clearpage`	Break and place pending floats first.

The requested length must include the heading and its spacing. The command does not measure the following block or control floating figures.

3.5 nobreaks – Keep a block together

This custom environment puts its contents in one indivisible vertical box. If the block does not fit in the space remaining on the page, LaTeX moves the whole block to the next page. Use it for a short explanation, a heading with its paragraph, or a group of equations that should stay together.

Definition

Already defined in `Packages/Layout/usepackages.tex`, which is loaded by the template. In another project, add this definition once in the preamble. No additional package is required.

```
\newenvironment{nobreaks}{\vbox\bgroup}{\egroup}
```

Usage

Start and end the environment between paragraphs, with a blank line before and after the block. The example uses `lipsum` for placeholder text and `amsmath` for the equation.

```
\begin{nobreaks}
    \textbf{Local model}

    \lipsum[1][1-2]
    \[
        F = ma, \qquad a = \frac{F}{m}
    \]
    \lipsum[2][1]
\end{nobreaks}
```

Local model

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis.

$$F = ma, \qquad a = \frac{F}{m}$$

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi.

When replacing needspace

`\Needspace{length}` reserves a length before the following content, which may still split if it exceeds that reservation. `nobreaks` encloses the actual content and prevents all page breaks inside it; there is no length to estimate.

- The entire block must fit on one page. A taller block cannot split or shrink automatically and can run beyond the page boundary.
- Keep ordinary floating `figure` and `table` environments outside the box. Inside it, use non-floating content such as `\includegraphics` or `tabular`.
- Moving the block may leave unused space at the bottom of the preceding page.

4 Mathematical typesetting packages

4.1 amsmath – Equations and matrices

Preamble

```
\usepackage{amsmath}
```

Aligned equations – one number per line

```
\begin{align}
    F &= ma \label{eq:dynamics} \\
    a &= \frac{F}{m} \label{eq:acceleration}
\end{align}
```

$$F = ma \tag{2}$$

$$a = \frac{F}{m} \tag{3}$$

Several lines – one equation number

```
\begin{equation}
    \begin{split}
        E &= T + V \\
          &= \frac{1}{2}mv^2 + mgh
    \end{split}
    \label{eq:energy}
\end{equation}
```

$$\begin{aligned} E &= T + V \\ &= \frac{1}{2}mv^2 + mgh \end{aligned} \tag{4}$$

Use `align*` for no numbering, `\notag` to suppress one line number, and `\eqref{label}` for a reference.
`&` marks alignment;
`\\` separates lines.

Matrices and piecewise functions

```
\[
    A = \begin{bmatrix}
        1 & 2 \\
        3 & 4
    \end{bmatrix}
    \qquad
    \det(A) = \begin{vmatrix}
        1 & 2 \\
        3 & 4
    \end{vmatrix} = -2
\]
```

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad \det(A) = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -2$$

`pmatrix`: parentheses; `bmatrix`: brackets; `vmatrix`: determinant bars; `matrix`: no delimiters.

```
\[
    f(x) = \begin{cases}
        x^2, & \text{if } x \geq 0, \\
        -x,  & \text{if } x < 0.
    \end{cases}
\]
```

$$f(x) = \begin{cases} x^2, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

```
\[
    \left(\frac{a+b}{c}\right)^2
    \qquad \sin(\theta)
    \qquad \mathbf{v} = (v_x,v_y)
\]
```

$$\left(\frac{a+b}{c}\right)^2 \qquad \sin(\theta) \qquad \mathbf{v} = (v_x, v_y)$$

Use `\text{...}` for words and commands such as `\sin` or `\log` for operators.

4.2 cancel – Cancellations

Preamble

```
\usepackage{amsmath,cancel}
```

Common factors

```
\[
    \frac{\cancel{x}y}{\cancel{x}z}
    = \frac{y}{z},
    \qquad x \ne 0,\; z \ne 0
\]
```

$$\frac{\cancel{x}y}{\cancel{x}z} = \frac{y}{z}, \qquad x \neq 0,\ z \neq 0$$

Cancel towards a value

```
\[
    a + \cancelto{0}{(b-b)} = a
\]
```

$$a + \cancelto{0}{(b-b)} = a$$

Stroke styles

```
\[
    \cancel{x} \qquad \bcancel{x} \qquad \xcancel{x}
\]
```

$$\cancel{x} \qquad \bcancel{x} \qquad \xcancel{x}$$

4.3 siunitx – Numbers and units

Preamble

```
\usepackage{siunitx}
\sisetup{
    output-decimal-marker={.},
    per-mode=symbol
}
```

Quantities, numbers and angles

```
\[
    v = \qty{25}{\meter\per\second}
    \qquad
    a = \qty{9.81}{\meter\per\second\squared}
\]
\[
    \rho = \qty{1.225}{\kilogram\per\meter\cubed}
    \qquad F = \qty{15}{\newton}
\]
Numbers: \num{12500} and \num{3.2e8}.
Units only: \unit{\pascal} and \unit{\newton\meter}.
Angles: \ang{30} and \ang{45;30;0}.
```

$$v = 25\,\mathrm{m/s} \qquad a = 9.81\,\mathrm{m/s^2}$$

$$\rho = 1.225\,\mathrm{kg/m^3} \qquad F = 15\,\mathrm{N}$$

Numbers: 12 500 and 3.2×10^8. Units only: Pa and N m. Angles: 30° and 45°30′0″.

Set `output-decimal-marker={,}` for a decimal comma and `per-mode=reciprocal` for negative unit powers. `\qty` and `\unit` require `siunitx` v3; the older v2 equivalents are `\SI` and `\si`.

5 Text formatting packages

5.1 lipsum – Placeholder text

Preamble

```
\usepackage{lipsum}
```

Paragraphs and selected sentences

```
\lipsum[1]          % One complete paragraph
\lipsum[1-2][1]     % Paragraph range with sentence selection
\lipsum[1][1-3]     % First three sentences of paragraph 1
\lipsum*[2][1-3]    % No final paragraph break
```

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus.

Use this placeholder text to test columns, image wrapping and page layout. The first optional argument selects paragraphs; the second selects sentences. The starred form suppresses the paragraph break at the end of the generated text.

Reserved characters

```
Percentage: 25\%. Cost: 10\$. Identifier: case\_01.
Characters: \& \# \{ \}.
Backslash: \textbackslash{}. Tilde: \textasciitilde{}.
Caret: \textasciicircum{}.
```

Percentage: 25%. Cost: 10$. Identifier: case_01. Characters: & # { }. Backslash: \. Tilde: ~. Caret: ^.